Topic 3 Functions

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3.1 Preliminaries

Definition 3.1

A <u>function f</u> consists of:

- a nonempty set A called the <u>domain of f</u>;
- a nonempty set B called the <u>codomain of f</u>;
- and a set of ordered pairs $(a, b) \in (A, B)$ which assigns to each element of A an element of B.
- When f is a function with domain A and codomain B we write $f : A \longrightarrow B$

3.1 Preliminaries

Example 3.2

Is $g\colon \mathbb{R} \longrightarrow \mathbb{R}$ given by $g(x) = \dfrac{x+1}{x^2+4}$ a function?

3.1 Preliminaries

Example 3.3

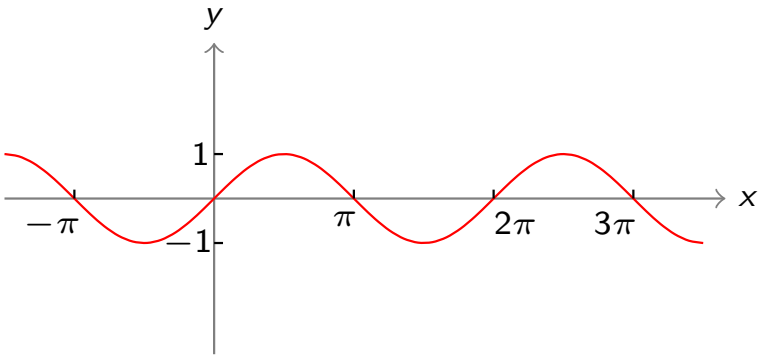
Explain why $h\colon \mathbb{C} \longrightarrow \mathbb{C}$ given by

$$h(z) = \frac{z+1}{z^2+4}$$

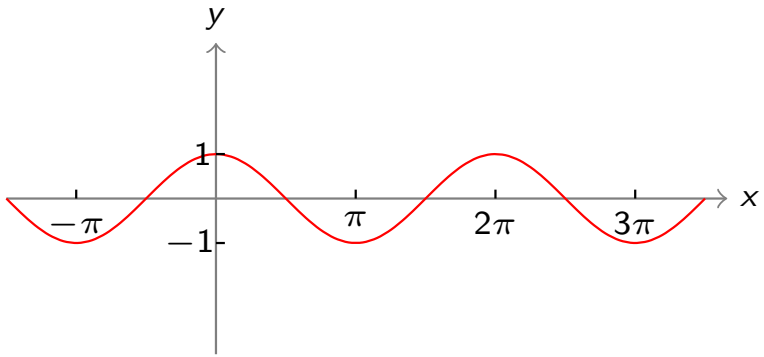
is not a function. How can we turn it into a function?

3.1 Preliminaries

- For a function f, when the domain and codomain are subsets of $\mathbb{R}$, we can also visualise it in the plane by plotting the pairs $(a, f(a))$.
- For instance, recall the graphs of the basic trigonometric functions:



$\sin\colon \mathbb{R} \longrightarrow \mathbb{R}$



$\cos\colon \mathbb{R} \longrightarrow \mathbb{R}$

3.1 Preliminaries: Image and Range

Definition 3.4

Let A and B be sets and let $f : A \longrightarrow B$. The <u>range</u> of f is the subset of elements of B to which elements of A actually map. We denote the range of f as $range(f)$

$$\text{range}(f) = \{b \in B \mid \text{there exists } a \in B \text{ such that } f(a) = b\}$$

$$\text{range}(f) = \{f(a) \mid a \in A\}.$$

3.1 Preliminaries: Image and Range

3.1 Preliminaries: Image and Range

We can also consider the set of values that a function can take for a restricted input.

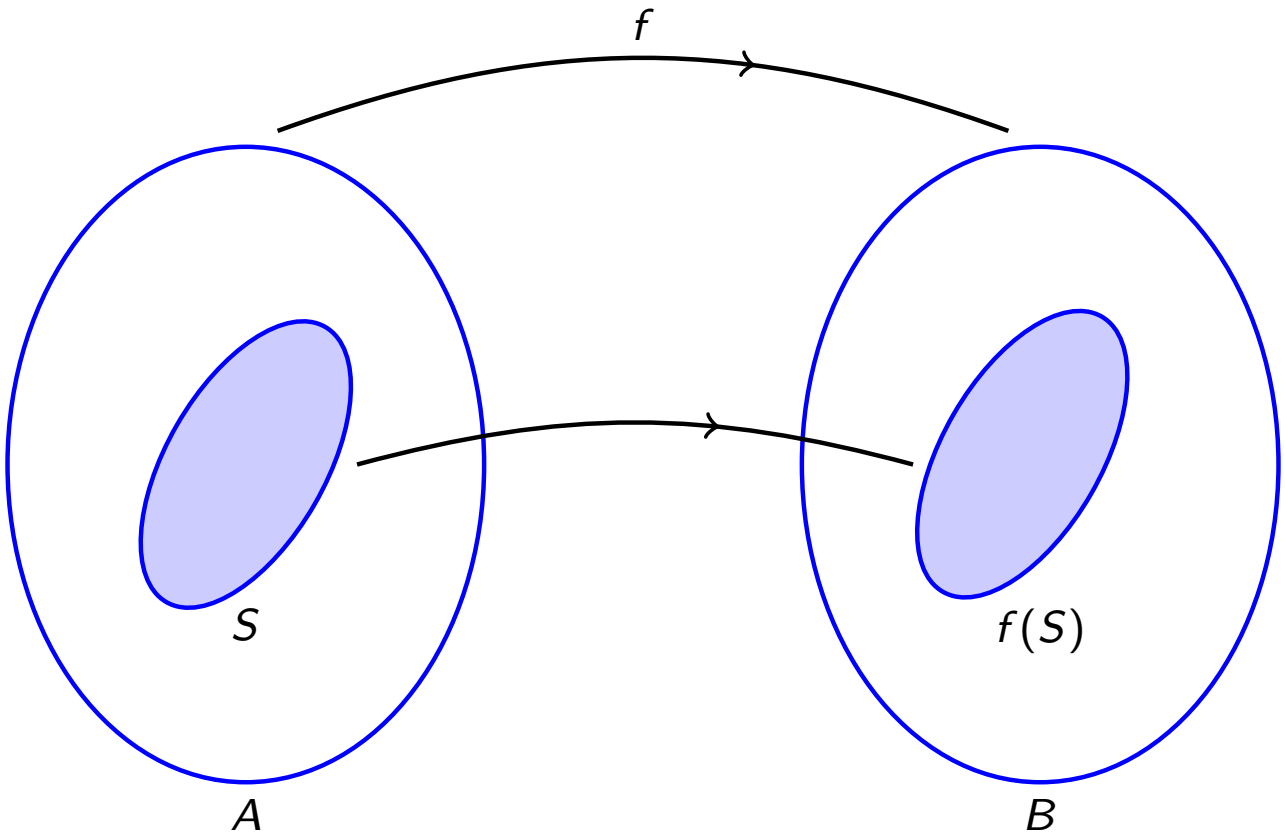
Definition 3.5

Let A and B be sets and let $f : A \longrightarrow B$. Let S be a subset of A. The <u>image of S under f</u> is the set

$$f(S) = \{b \in B \mid \text{there exists } s \in S \text{ such that } f(s) = b\}$$

$$f(S) = \{f(s) \mid s \in S\}.$$

In pictures



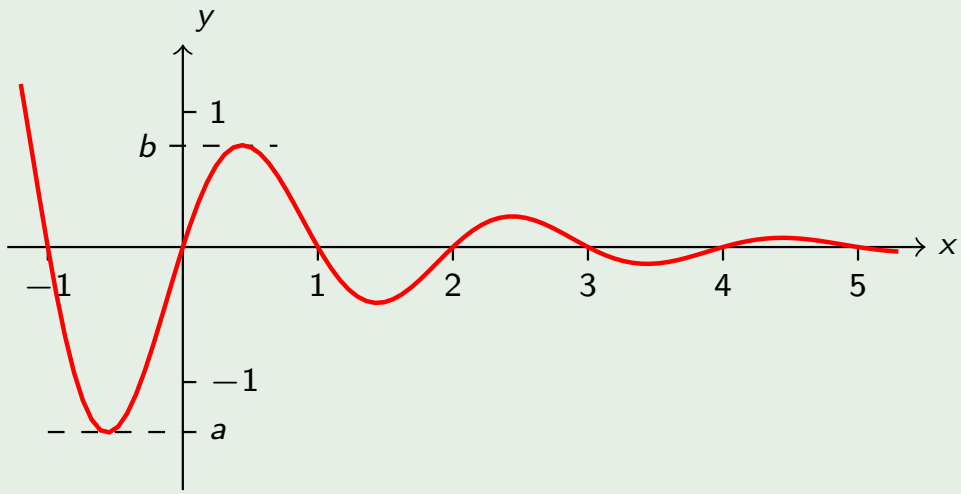
3.1 Preliminaries: Image and Range

3.1 Preliminaries

Example 3.6

For $f\colon \mathbb{R} \longrightarrow \mathbb{R}$ given by the graph, find:

1. $f([0,1])$.
2. $f((-\infty,-1))$.
3. $f([-1,2])$.
4. range(f).



(Assume the graph approaches ∞ on the left and oscillates toward 0 on the right.)

3.1 Preliminaries

Homework 27

Find the image of the interval $(-\frac{\pi}{6}, \pi]$ *under the function* $f : \mathbb{R} \longrightarrow \mathbb{R}$ *given by* $f(x) = \sin(x) + 1$.

3.1 Preliminaries: Non-numerical functions

- There is no reason to restrict the domain and codomain of a function to be sets of numbers.

Example 3.7

Let W denote the set of all English words and A the Roman alphabet. Consider $\alpha : W \longrightarrow A$ given by

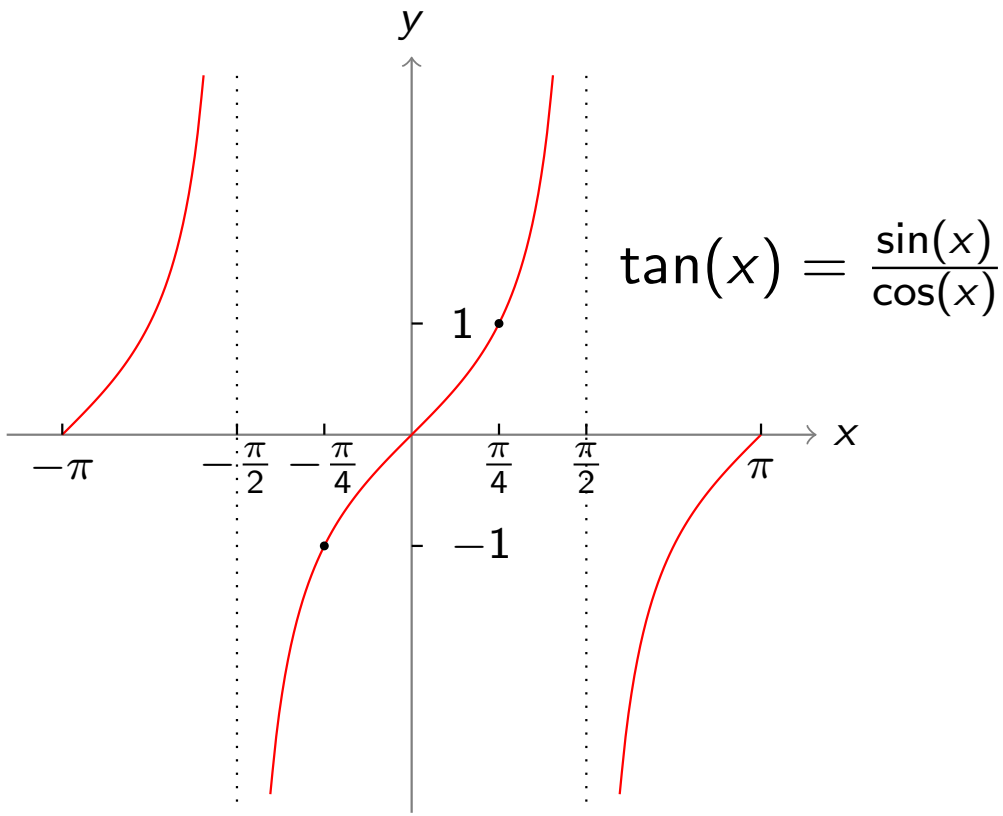
$$\alpha(w) = \text{the first letter in } w.$$

3.1 Preliminaries: Trigonometric Functions

- We already reviewed the graphs of the trigonometric functions sine and cosine. Their quotient:

$$\tan\colon \mathbb{R} \setminus \{(2k+1)\,\tfrac{\pi}{2} \mid k \in \mathbb{Z}\} \longrightarrow \mathbb{R}$$

called the <u>tangent function</u>, is given by $\tan(x) = \frac{\sin(x)}{\cos(x)}$



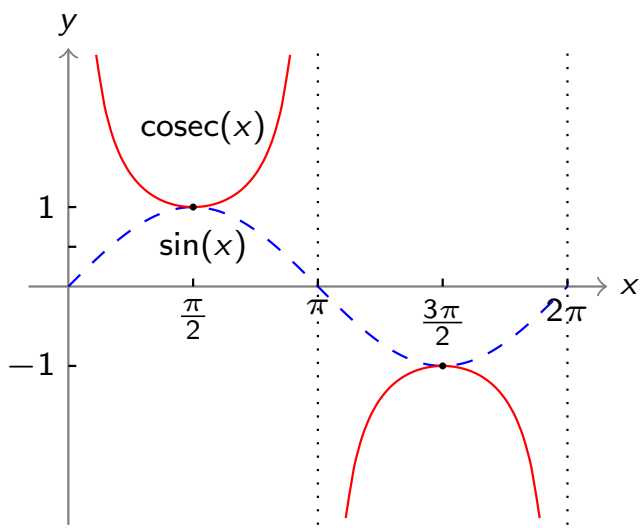
3.1 Preliminaries: Trigonometric Functions

Homework 28

Find the domain and sketch the graph of the <u>*secant*</u> *function* $\sec(x) = \dfrac{1}{\cos(x)}$.

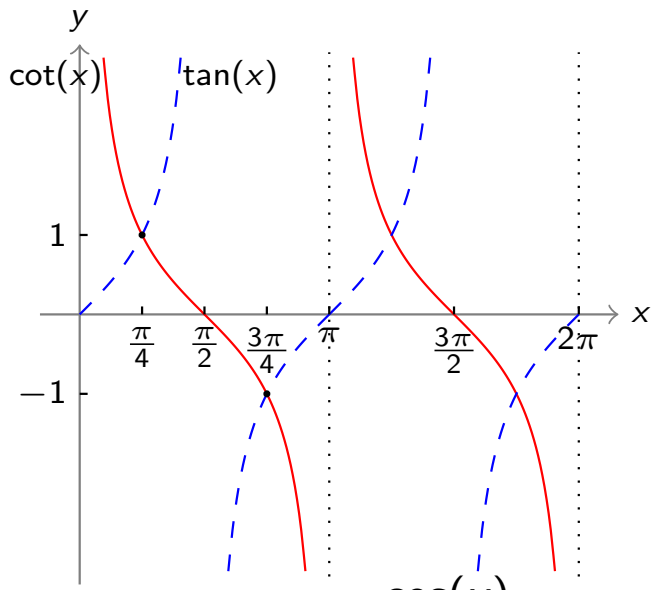
3.1 Preliminaries: Reciprocal trigonometric functions

- It is also useful to define and study the reciprocals of sine, cosine and tangent:



$$\operatorname{cosec}(x) = \frac{1}{\sin(x)}$$

Cosecant function



$$\cot(x) = \frac{\cos(x)}{\sin(x)}$$

Cotangent function

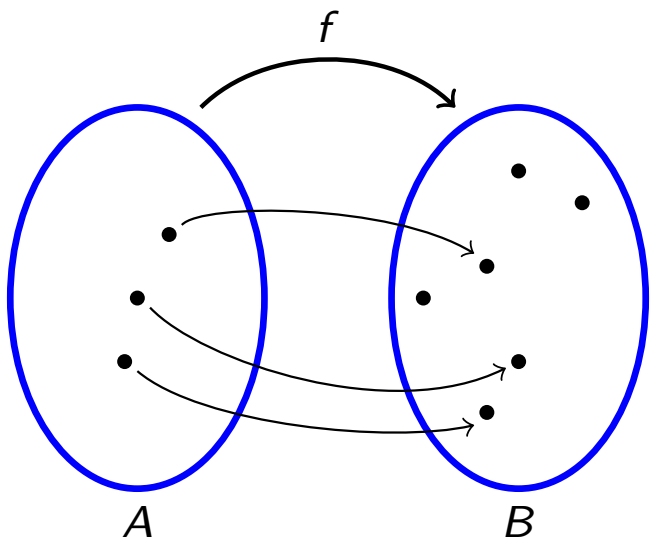
3.2 Injections, Surjections and Bijections

3.2 Injections, Surjections and Bijections: Injections

Definition 3.8

Let A and B be sets. Let $f : A \longrightarrow B$. We say f is <u>injective</u> when the equality $f(a_1) = f(a_2)$ implies $a_1 = a_2$ for every pair $a_1, a_2 \in A$

- Equivalently, injectivity means that $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$.
- In other words, f keeps the elements of A separated: no two distinct elements of A map to the same in the codomain.



3.2 Injections, Surjections and Bijections: Proving Injectivity

Example 3.9

Prove $g : \mathbb{R} \setminus \{1\} \longrightarrow \mathbb{R}$ defined by $g(x) = \dfrac{2x-7}{1-x}$ is injective.

Strategy: To prove a function g is injective we must deduce the following: for every $x_1, x_2 \in \mathrm{dom}(g)$, if $g(x_1) = g(x_2)$ then $x_1 = x_2$.

3.2 Injections, Surjections and Bijections: Proving Injectivity

Proof.

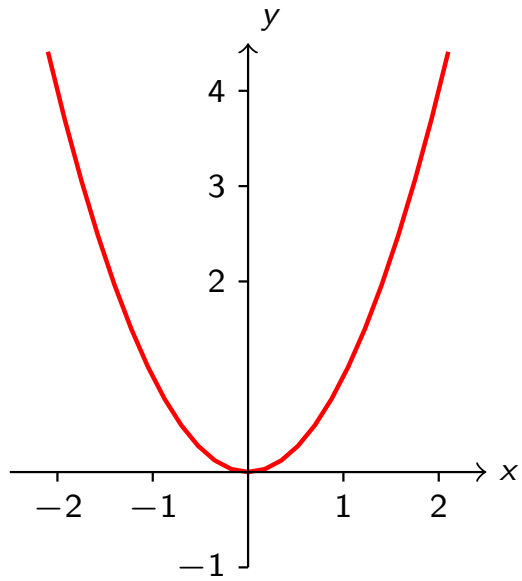
- Let $g : \mathbb{R} \setminus \{1\} \longrightarrow \mathbb{R}$ defined by $g(x) = \dfrac{2x-7}{1-x}$.
- Consider $x_1, x_2 \in \mathrm{dom}(g)$.
- Assume $g(x_1) = g(x_2)$

$\square$

3.2 Injections, Surjections and Bijections: Injective Functions

Example 3.10

Consider $f : \mathbb{R} \longrightarrow \mathbb{R}$ given by $f(x) = x^2$.
Is f injective? Find the largest interval(s) on which f is injective.



3.2 Injections, Surjections and Bijections: Injective Functions

3.2 Injections, Surjections and Bijections: Monotone Functions

Definition 3.11

Let A and B be subsets of $\mathbb{R}$ and let $f : A \longrightarrow B$. We say f is <u>monotone increasing</u> when $x_1 < x_2$ always implies $f(x_1) < f(x_2)$. We say f is <u>monotone decreasing</u> when $x_1 < x_2$ always implies $f(x_1) > f(x_2)$.

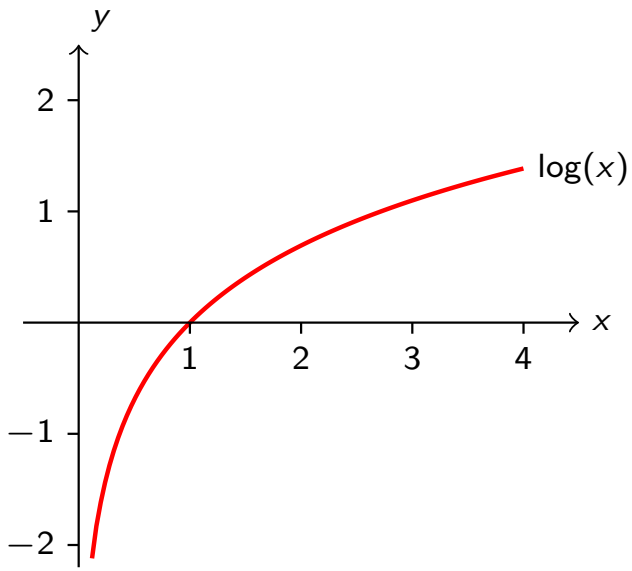
Theorem 3.12

Let A and B be subsets of $\mathbb{R}$ and let $f : A \longrightarrow B$. If f is monotone increasing or decreasing, then f is injective.

3.2 Injections, Surjections and Bijections: Monotone Functions

Example 3.13

Using the picture and the theorem explain how you know $\log\colon (0,\infty) \longrightarrow \mathbb{R}$ is injective.



3.2 Injections, Surjections and Bijections: Surjective Functions

- Recall the definition of range of a function:

- Necessarily, $\text{range}(f) \subseteq B$. It is certainly possible to have $\text{range}(f) = B$.

3.2 Injections, Surjections and Bijections: Surjective Functions

Definition 3.14

Let A and B be sets and let $f : A \longrightarrow B$. We say f is <u>surjective</u> when $range(f) = B$.

- In other words, for every $b \in B$ there exists at least one $a \in A$ such that $f(a) = b$.

3.2 Injections, Surjections and Bijections: Proving Surjectivity

Example 3.15

Prove $f : \mathbb{R} \setminus \{-1\} \longrightarrow \mathbb{R} \setminus \{2\}$ defined by

$$f(x) = 2 - \frac{1}{x+1}$$

is surjective.

Strategy: Consider an arbitrary element of the codomain. Deduce there is an element of the domain that maps to that element of the codomain.

3.2 Injections, Surjections and Bijections: Proving Surjectivity

Proof.

- Consider $y \in \mathbb{R} \setminus \{2\}$.
- We seek $x \in \mathbb{R} \setminus \{-1\}$ such that $f(x) = y$.
- Substituting in the function f we have $2 - \frac{1}{x+1} = y$

$\square$

3.2 Injections, Surjections and Bijections: Bijections

3.2 Injections, Surjections and Bijections: Bijections

Definition 3.16

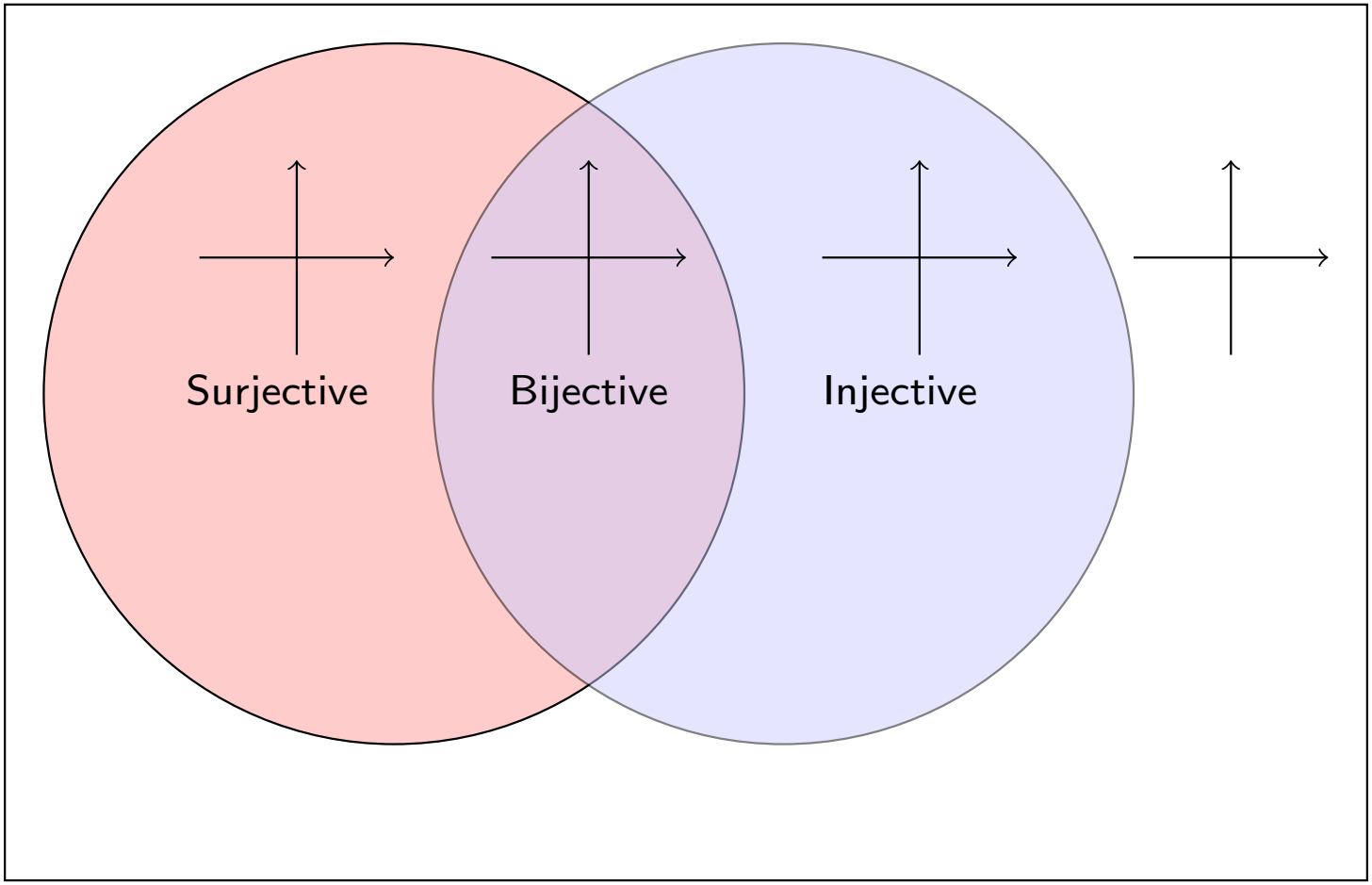
Let A and B be sets and let $f : A \longrightarrow B$. We say f is <u>bijective</u> when it is injective and surjective.

- A bijective function f gives a complete correspondence between the elements of A and the elements of B.

3.2 Injections, Surjections and Bijections

Example 3.17

Give examples of functions $\mathbb{R} \longrightarrow \mathbb{R}$ in every part of the Venn diagram.

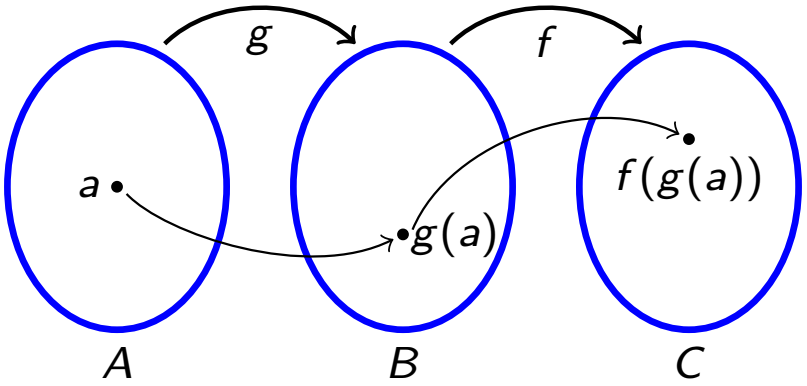


3.3 Composition and Inverse: Building New Functions from Old

- We can build up complicated function formulas using combinations of algebraic operations.
- Letting $f(x) = x^2, g(x) = e^{-x}, h(x) = \cos(x)$:
  - **Addition and subtraction:** $f(x) + g(x) =$
  - **Multiplication and division:** $f(x)g(x) =$
  - **Multiplying by a constant:** $-5h(x) =$
- There is one more method we use to combine known functions to obtain new ones - function composition.

3.3 Composition and Inverse: Function Composition

Idea: What happens if we build a function by using the output of one function as the input for the next function?



3.3 Composition and Inverse: Function Composition

Example 3.18

Let $f, g : \mathbb{R} \longrightarrow \mathbb{R}$ be respectively given by $f(x) = 2x + 1$ and $g(x) = x^2 + 3$. Find formulas for $f(g(x))$ and $g(f(x).)$

3.3 Composition and Inverse: Function Composition

Example 3.19

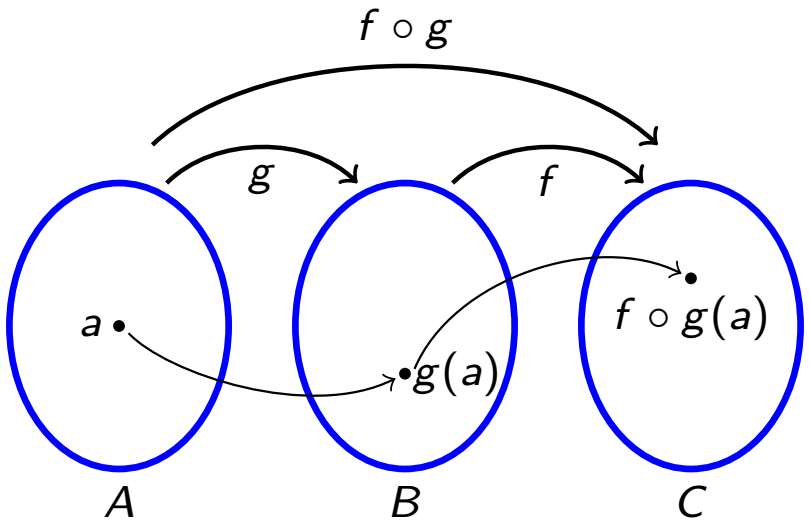
Let $f, g : \mathbb{R} \longrightarrow \mathbb{R}$ be respectively given by $f(x) = 2x + 1$ and $g(x) = \frac{x-1}{2}$. Find formulas for $f(g(x))$ and $g(f(x))$.

3.3 Composition and Inverse

Definition 3.20

Let A, B and C be sets and let $g : A \longrightarrow B$ and $f : B \longrightarrow C$. The <u>composition of f and g</u> is defined to be the function $f \circ g : A \longrightarrow C$, where

$$f \circ g(a) = f(g(a))$$



3.3 Composition and Inverse

Homework 29

Consider $f, g : \mathbb{C} \longrightarrow \mathbb{C}$ *given by*

$$f(z) = i$$
$$g(z) = z.$$

Compare $f \circ g$ *to* $g \circ f$.
Note: $f(z)$ *is an* <u>*constant*</u> *function. It gives the same output for every input.*

3.3 Composition and Inverse: Domain and Range

Question: What happens to the domain and range when we compose functions?

- Let $f(x) = \log(x)$ and $g(x) = 1 - x^2$

3.3 Composition and Inverse: Domain and Range

- Suppose we are given functions $f : A \longrightarrow B$ and $g : C \longrightarrow D$ and we want to find $\mathrm{dom}(f \circ g)$:
  1. To perform the first calculation to get $g(x)$, we must have $x \in \mathrm{dom}(g)$.
  2. To complete the calculation of $f(g(x))$, we next need $g(x) \in \mathrm{dom}(f)$.
- We can express these two requirements in set-builder notation. This is the <u>implied domain</u>.

$$\mathrm{dom}(f \circ g) = \{x \in \mathrm{dom}(g) \mid g(x) \in \mathrm{dom}(f)\}.$$

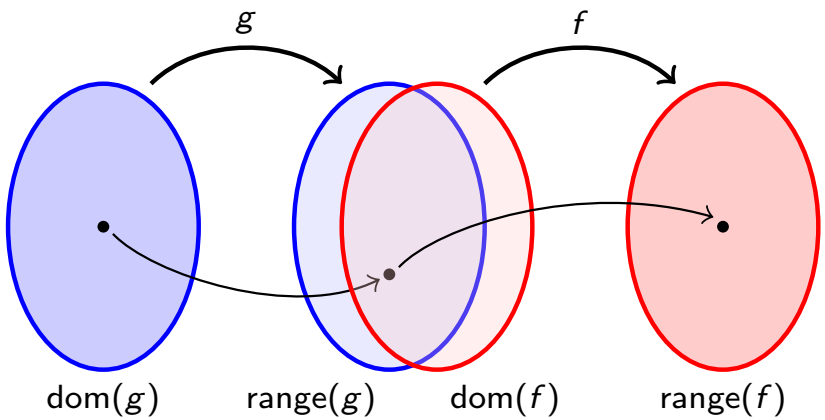
3.3 Composition and Inverse: Domain and Range of Compositions

- For any x in the set

$$D = \mathrm{dom}(f \circ g) = \{x \in \mathrm{dom}(g) \mid g(x) \in \mathrm{dom}(f)\},$$

  $g(x) \in \mathrm{range}(g)$ by definition of $\mathrm{range}(g)$ and $g(x) \in \mathrm{dom}(f)$ by definition of D. This gives $g(x) \in \mathrm{range}(g) \cap \mathrm{dom}(f)$.
- The <u>implied range</u> of $f \circ g$ is the set

$$\mathrm{range}(f \circ g) = f\big(\mathrm{range}(g) \cap \mathrm{dom}(f)\big).$$



3.3 Composition and Inverse: Domain and Range of Compositions

Back to our example:

Example 3.21

Find the implied domain and range of

$$f \circ g(x) = \log(1 - x^2)$$

$$\begin{aligned} \operatorname{dom}(f \circ g) &= \{x \in \operatorname{dom}(g) \mid g(x) \in \operatorname{dom}(f)\} \\ &= \end{aligned}$$

$$\begin{aligned} \operatorname{range}(f \circ g) &= f\left(\operatorname{range}(g) \cap \operatorname{dom}(f)\right) \\ &= \end{aligned}$$

3.3 Composition and Inverse: Domain and Range of Compositions

Example 3.22

1. Find the implied domain and range of the expression $\sqrt{\sin(x)}$.
2. Find the implied domain of $\sqrt{\cos(x) - 2}$.

3.3 Composition and Inverse: Inverse

Definition 3.23

Let A and B be sets and let $f : A \longrightarrow B$. Let $g : B \longrightarrow A$. We say <u>g is the inverse of f</u> when

$$g \circ f(x) = x$$

and

$$f \circ g(y) = y$$

for every $x \in A$ and every $y \in B$. When g is the inverse of f we may refer to g as f^{-1}.

3.3 Composition and Inverse: Inverse

- If f has an inverse, then no matter which input we put in we must be able to *reverse it*

3.3 Composition and Inverse: Inverse

Theorem 3.24

Let A and B be sets and let $f : A \longrightarrow B$

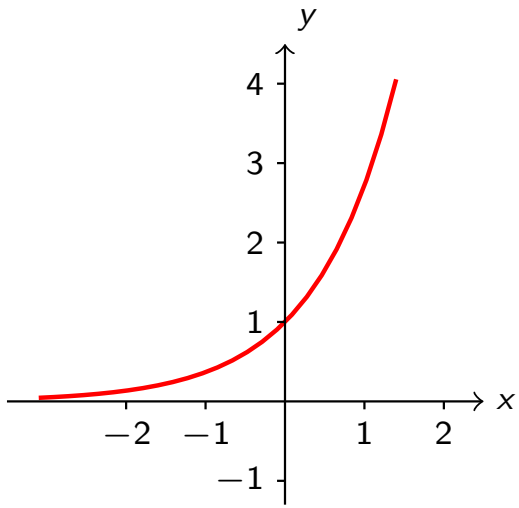
1. *The function f has an inverse if and only if f is bijective*
2. *The inverse of f is unique.*

Aside: If and only if statements

The function <u>f has an inverse</u> if and only if <u>f is bijective</u>

3.3 Composition and Inverse: Familiar Examples

- Question: does $f(x) = e^x$ have an inverse?
- Answer: ... it depends..



3.3 Composition and Inverse: Familiar Examples

Example 3.25

Using Definition 3.23 , prove $g\colon \mathbb{R} \longrightarrow \mathbb{R}$ given by $g(x) = \frac{x+4}{3}$ is the inverse of $f\colon \mathbb{R} \longrightarrow \mathbb{R}$ given by $f(x) = 3x - 4$

3.3 Composition and Inverse: Why are Inverses Helpful?

Example 3.26

Solve the equation $e^{x^3+1} = e^3$ for $x \in \mathbb{R}$.

3.3 Composition and Inverse: Properties of Invertible Functions

Example 3.27

Does the function $f : \mathbb{C} \longrightarrow \mathbb{C}$ defined by $f(z) = \bar{z}$ have an inverse? Is it bijective?

3.3 Composition and Inverse: Properties of Invertible Functions

Question: is $f(x) = x^2$ invertible?
Answer:

3.3 Composition and Inverse: Resolving non-surjectivity

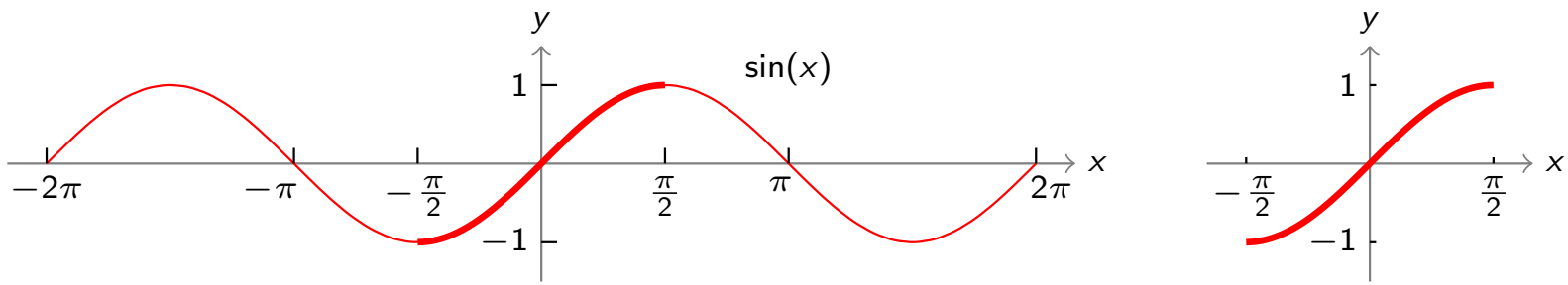
- There are a lot of interesting functions that are neither injective nor surjective.
- By the previous theorem, such functions do not have inverses.
- Functions that are not surjective can be modified to be surjective by replacing the codomain with the range.
- In the rest of this section we may assume that our functions are surjective.

3.3 Composition and Inverse: Resolving non-injectivity

- To resolve non-injectivity, we have to restrict the function $f\colon A \longrightarrow B$ to some subset C of A on which f is injective.
- In other words, we define a function $f_C\colon C \longrightarrow B$ given by $f_C(c) = f(c)$ for all $c \in C$.
- This function is called the <u>restriction</u> of f to C.
- Then f_C is bijective and therefore has an inverse $g_C\colon B \longrightarrow C$.

3.3 Composition and Inverse: Inverse Trig Functions

- Notice, sine is not injective because it is periodic.



- Let us restrict the domain of sin to a subset (as big as possible) on which it is injective.
- There are infinitely many possible choices, but the convention is to restrict to $[-\frac{\pi}{2}, \frac{\pi}{2}]$.
- The range of this restricted function is still $[-1, 1]$, so $f : [-\frac{\pi}{2}, \frac{\pi}{2}] \longrightarrow [-1, 1]$.
- Since the function is now injective and surjective, then it is bijective. And so it has an inverse!

3.3 Composition and Inverse: Inverse Sine

Definition 3.28

Consider $\sin : [-\frac{\pi}{2}, \frac{\pi}{2}] \longrightarrow [-1, 1]$. The inverse of this function called <u>arcsin</u>.

$$\arcsin : [-1, 1] \longrightarrow [-\tfrac{\pi}{2}, \tfrac{\pi}{2}]$$

By definition

$$\sin \circ \arcsin(x) = x \quad \text{for all } x \in [-1, 1]$$

and

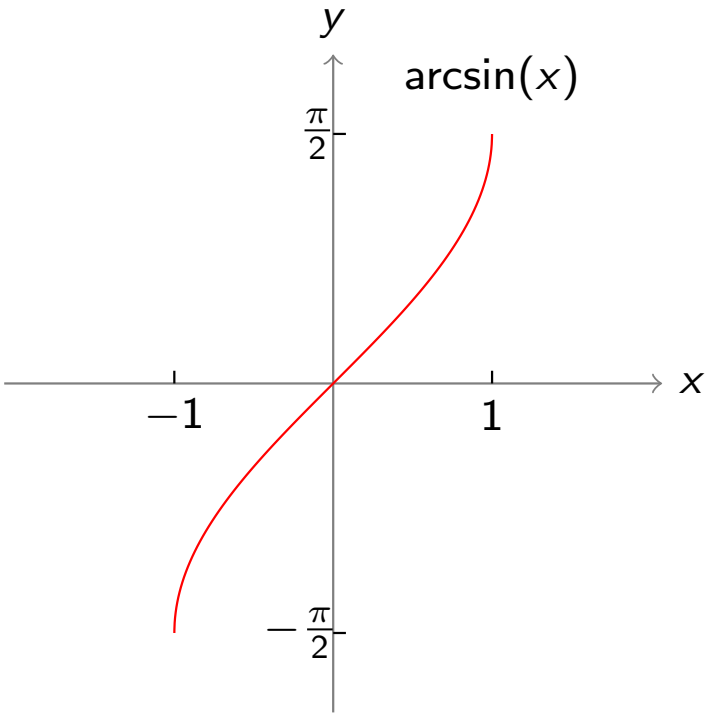
$$\arcsin \circ \sin(\theta) = \theta \quad \text{for all } \theta \in [-\tfrac{\pi}{2}, \tfrac{\pi}{2}]$$

3.3 Composition and Inverse: Inverse Sine

Definition 3.29

Consider $\sin : [-\frac{\pi}{2}, \frac{\pi}{2}] \longrightarrow [-1, 1]$. The inverse of this function called <u>arcsin</u>.

$$\arcsin : [-1, 1] \longrightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$$



3.3 Composition and Inverse: Notation for Inverse Trig Functions

- This function is also referred to on calculators (and in some texts) as Sin^{-1} where the capital S denotes the restricted domain and the superscript -1 means inverse function rather than reciprocal.
- In place of this, we opt for arcsin.
- This avoids potential confusion between $\mathrm{Sin}^{-1}(x)$ and $\frac{1}{\sin(x)}$.
- Similarly, we avoid $\mathrm{Cos}^{-1}(x)$ or $\mathrm{Tan}^{-1}(x)$.

3.3 Composition and Inverse: Properties of arcsin

1. $\arcsin(\sin(x)) = x$ provided $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$,
2. $\sin(\arcsin(y)) = y$ for any $y \in [-1, 1]$,
3. When we are solving equations, we can always use

$$\theta = \arcsin(y) \Rightarrow \sin(\theta) = y$$

4. We can only use the argument

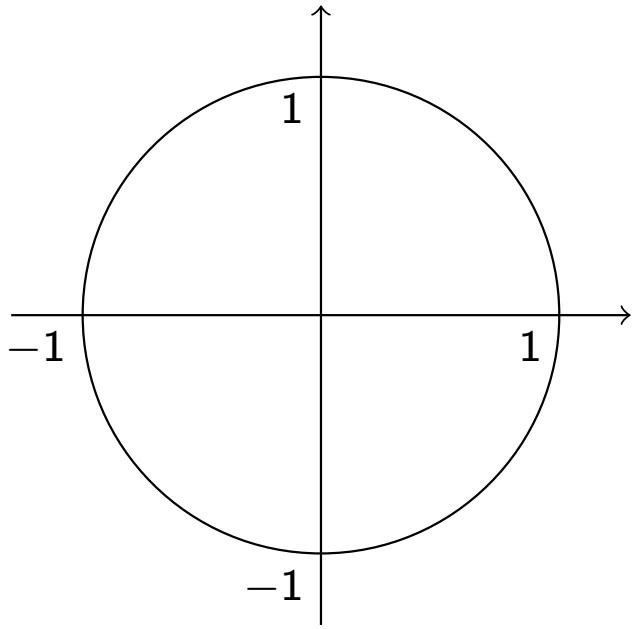
$$\sin(x) = y \Rightarrow x = \arcsin(y)$$

if we know for sure $x \in \text{range}(\arcsin) = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

3.3 Composition and Inverse: Properties of arcsin

Example 3.30

Simplify $\arcsin\left(\sin\left(\frac{3\pi}{4}\right)\right)$.



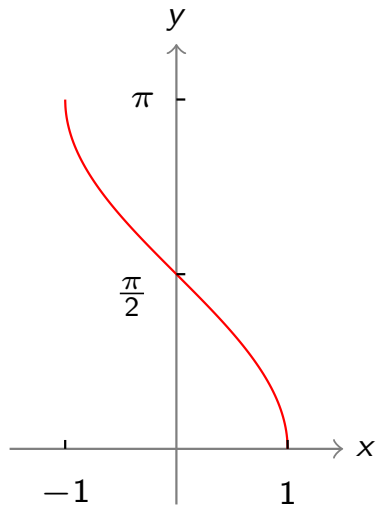
3.3 Composition and Inverse: Properties of arcsin

Homework 30

Evaluate $(\arcsin \circ \sin)\left(\frac{2\pi}{3}\right)$

3.3 Composition and Inverse: Properties of arccos, arctan

- We can perform the same analysis with cos and tan. Appropriately restricted, they have the following inverses:

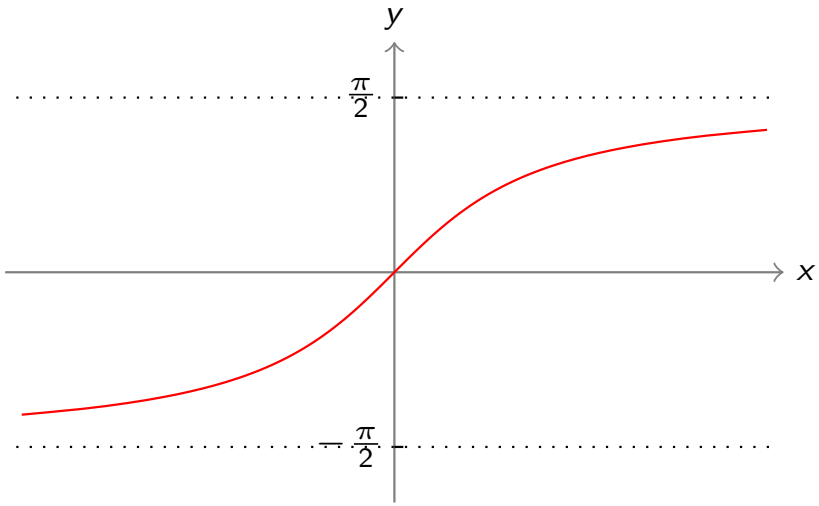


$$\arccos : [-1, 1] \longrightarrow [0, \pi]$$

- $\cos \circ \arccos(x) = x$ for all $x \in [-1, 1]$.
- $\arccos \circ \cos(\theta) = \theta$ for all $\theta \in [0, \pi]$.

3.3 Composition and Inverse: Properties of arccos, arctan

- We can perform the same analysis with cos and tan. Appropriately restricted, they have the following inverses:



$$\arctan : \mathbb{R} \longrightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

- $\tan \circ \arctan(x) = x$ for all $x \in \mathbb{R}$.
- $\arctan \circ \tan(\theta) = \theta$ for all $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

3.3 Composition and Inverse: Domain and Range of Compositions

Homework 31

Find the implied domain and range of the expression

$$\arcsin\left(\sqrt{2}\,\sin(x)\right)$$